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SentiEntityRec: Entity-level sentiment perception graph neural network for news recommendation

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Abstract
Precise news recommendations are critical in today's digital landscape. However, conventional approaches overlook fine-grained sentiment nuances associated with individual entities in news content. This paper presents SentiEntityRec, a novel graph neural network framework that enriches traditional entity embeddings using a global graph-enhanced model by incorporating sentiment vectors. Experiments conducted with MIND datasets have shown that SentiEntityRec surpasses existing models in key performance metrics. For example, the proposed model improves +0.5% AUC over GLoCIM. These results underscore the superior efficacy of incorporating entity-sentiment analysis into graph-based news recommendation systems.

Keywords Sentiment Analysis · Graph Neural Networks · News Recommendation · Personalization

1 Introduction

In the digital age, the Internet has become an essential and omnipresent digital environment where people communicate, seek information, and make decisions (Kozyreva et al. 2020). Online news distribution platforms such as Microsoft News have garnered hundreds of millions of users (Wu et al. 2020b). The exponential growth in the volume of online news content has made it increasingly challenging for individuals to find articles that interest them (Wu et al. 2023a) efficiently. Amid this information overload, personalized news recommendation systems have become crucial tools to help users sort through vast amounts of news and discover content that matches their preferences (Wu et al. 2021a). Traditional recommendation methods, including collaborative and content-based filtering, have been widely used to tackle this issue. However, these methods often face limitations such as data sparsity and cold start problems, particularly in the fast-changing news domain (Feng et al. 2020).

Recent advances in natural language processing (NLP) and machine learning have led to the development of more sophisticated recommendation algorithms. Graph neural networks (GNNs) have particularly shown great promise in capturing complex relationships between users, news articles, and their underlying entities (Veličković et al. 2017). By representing news articles and user interactions as nodes and edges in a graph, GNNs can effectively propagate information and learn rich representations that account for both content and relational information (Kipf and Welling 2016).

Despite these advancements, existing GNN-based recommendation systems focus primarily on semantic and relational characteristics. They often neglect the sentiment context of news articles, leading to recommendations that, while relevant, may not fully resonate with the user's sentimental state or preferences. This potentially diminishes user satisfaction and engagement (Ding et al. 2023). Studies have shown that users' sentiment responses can significantly impact the speed, breadth, and depth of information spread (Zhang et al. 2017). Sentiment analysis, which involves detecting and extracting subjective information from text, is a powerful tool for understanding user preferences and enhancing recommendation accuracy (Pang et al. 2008). So, efficiently filtering and delivering content that aligns with users' interests and sentiment states has become an urgent

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challenge. Therefore, News Sentiment Classification and Recommendation (NSCR) systems have become essential. They assist users in navigating the vast sea of content by delivering articles that match their preferences and interests (Xiao et al. 2021).

Reis, Olmo Benevenuto, Prates, and An introduced a method to uncover the correlation between sentiment polarity and news popularity (Rieis et al. 2015). Their study revealed that the sentiment polarity of headlines significantly influences the popularity of news articles. Specifically, they found that headlines with negative or positive sentiments tend to attract more attention than those with a neutral tone.

Shirsat, Jagdale, and Deshmukh presented a method for conducting sentiment analysis at the document level, enabling the assessment of the overall sentiment polarity of an entire news article (Shirsat et al. 2017). Document-level sentiment analysis is effective for gauging the general sentiment of a news piece. But it falls short in the context of news recommendations when a single article encompasses multiple entities, each with distinct sentiment polarities. For example:

"*Alibaba reported record-breaking Singles' Day sales, exceeding analysts' expectations. Meanwhile, Tencent announced a decline in advertising revenue due to regulatory changes. Xiaomi launched its new flagship smartphone, receiving mixed reviews from users*"

The document-level sentiment analysis (Loureiro et al. 2022) output polarity is neutral. The positive sentiment from Alibaba's sales is mixed with negative sentiment from Tencent's revenue decline and mixed reactions to Xiaomi's smartphone launch. If Tencent investors read this news, they may perceive it as negative, while Alibaba stakeholders may find it positive. Rønningstad et al. (2023) reveals that merely 47.7% of entities analyzed through entity-level sentiment analysis share the same polarity as their corresponding document-level labels.

Sentence-level sentiment analysis evaluates each sentence separately, avoiding the problem of aggregating different entity sentiments. However, it still struggles with sentences containing multiple entities with contrasting sentiment polarities. For example:

"*Alibaba and Tencent released financial reports, but Tencent's advertising revenue declined, while Alibaba's e-commerce growth remained strong*"

The model (Sanh et al. 2019) classifies this sentence as positive, even though Tencent is associated with negative sentiment. Sentence-level sentiment analysis cannot assign sentiment to specific entities within a sentence. By incorporating sentiment information at the entity level, we can gain deeper insights into the sentiment tone of news articles and how they influence user engagement. According to Rønningstad et al. (2023), inferring entity-level sentiment from sentence sentiment can correctly resolve the sentiment of 69.5% of entities. While this outperforms document-level sentiment analysis, it is still not sufficient.

Entity-level sentiment analysis can address the limitations of document-and sentence-level approaches by determining the sentiment expressed toward each entity separately, even within the same sentence. For the previous example, entity-level sentiment analysis outputs that Alibaba is positive and Tencent is negative. Thus, entity-level sentiment analysis can capture the sentiment for each entity.

In the context of news recommendation systems, consider a scenario where a recommendation will be generated for user $User_k$, as depicted in Fig. 1. In instances where the candidate news has the same entities in the news reading history of user $User_k$, recommend news to $User_k$. Traditional recommendation algorithms typically recommend candidate news items n_i, n_j and n_k to $User_k$ based on the same entities of the news. However, our model will achieve a distinct recommendation outcome by incorporating the sentiment-aware entity proposed in this study.

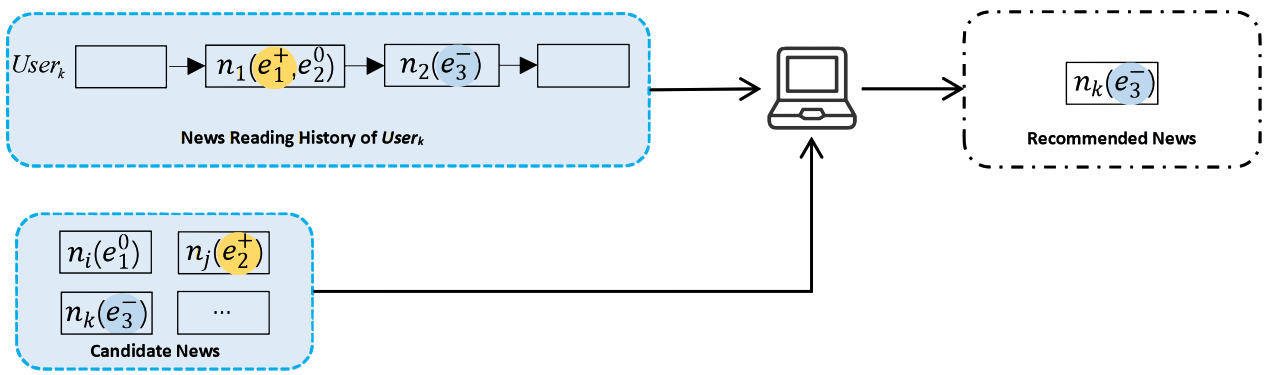


Fig. 1 An example of how the sentiment perception entity graph works to predict the candidate news. $User_k$ represents a user. n_1, n_2, n_i, n_j and n_k represent news articles. Representations in parentheses after the news are the entities of the news. The symbols "+", "0", and "-" at the top right of the entity indicate the "Positive", "Neutral", and "Negative" polarity of the entity, respectively

Through entity-level sentiment analysis, we can obtain the sentiment polarity of each entity in the news. The entities in candidate news n_i, n_j and n_k appear in the user history reading news n_1 and n_2, but their sentiment polarities are not the same. Only entity e_3 in candidate news n_k and entity e_3 in user history reading news e_2 are the same in terms of semantics and sentiment polarity. Because of the dimension of the sentiment polarity of the entities considered, we can find that the recommendation priority of candidate news n_k is higher than that of candidate news n_i and n_j. This is precisely where this model differs from other news recommendation algorithms. This divergence underscores the enhanced precision and personalization afforded by the sentiment-aware entities in capturing nuanced user preferences and news item characteristics.

The advantages of entity-level sentiment analysis include: 1) Capture sentiment for each entity - Ensures that each entity's sentiment is analyzed independently; 2) Prevent misleading sentiment labels - A single article or sentence can have both positive and negative sentiments; 3) Enhance recommendation accuracy - Users receive news aligned with their specific entity preferences rather than generalized sentiment scores.

Given the limitations of document- and sentence-level sentiment analysis, entity-level sentiment analysis is the most suitable approach for sentiment perception news recommendations. It ensures that 1) Users receive news recommendations that align with their entity-specific interests; 2) the sentiment vector of each entity is correctly captured, avoiding misleading classifications; and 3) Sentiment-aware recommendation systems can leverage entity embeddings with sentiment vector, improving personalization.

To solve those problems, we present SentiEntityRec. This novel framework aims to bridge the gap by incorporating sentiment vectors into Graph Neural Networks (GNNs) for enhanced news recommendation. Our approach captures semantic and relational aspects of news articles and integrates sentiment information to provide more personalized and context-aware recommendations. This work builds upon recent studies (Tien et al. 2024; Tao and Alatas 2024a) that have demonstrated the effectiveness of sentiment-aware recommendation systems and the potential of GNNs in handling complex relational data. The primary contributions of this work are highlighted as follows:

1. We propose the integration of the sentimental feature into the local entity encoding process within GNN-based recommendation systems
2. We design a local context-aware entity encoder that fuses the sentimental and semantic features of entities
3. We conduct comprehensive experiments on benchmark datasets, demonstrating that SentiEntityRec significantly outperforms state-of-the-art baselines

In the following sections of this paper, we introduce related works in Section 2. The methodology is presented in Section 3. A detailed description of the experiments is in Section 4. Section 5 concludes the paper and discusses future directions.

2 Related work

2.1 Personalized news recommendation systems

With the development of the Internet, more and more information has been moved to the Internet. News has also moved from traditional media to the Internet. News Aggregators[1] like Microsoft News[2], Google News[3], and Yahoo! News[4] appear. Initially, these platforms relied on recommendation methods based solely on non-personalized factors such as the popularity and recency of the news.

Personalized news recommendation(PNR) has emerged as a crucial technology aiming to deliver news articles that align with individual users' reading interests (Meng et al. 2023). With the growing volume of news and the increasing number of online readers, the importance of personalized news services has become more pronounced. Consequently, PNR algorithms have been developed and can be categorized into three main types: traditional PNR methods, deep learning-based methods, and graph structure learning-based methods.

The traditional news recommendation system has three filtering methods: content-based, collaborative, and hybrid filtering (Aggarwal et al. 2016). These three traditional methods place excessive reliance on manually engineered feature extraction.

In contrast, deep learning can automatically extract effective features from intricate content and characterize user and news representations through a nonlinear network. As a result, deep learning has found extensive application in news recommendation systems (LeCun et al. 2015). The wealth of textual information in news recommendation scenarios confers a substantial advantage. With the evolution of natural language processing (NLP) techniques, there has been a

[1] https://www.wpbeginner.com/showcase/best-news-aggregator-websites-how-to-build-your-own/

[2] https://news.microsoft.com/

[3] https://news.google.com/

[4] https://news.yahoo.com/

growing emphasis on content-based news recommendation systems. For example, NAML (Wu et al. 2019a) employs convolutional neural networks (CNNs) (Hughes et al. 2017) to capture news representations and gated recurrent units (GRUs) (Chung et al. 2014) to model user representations. Conversely, NRMS (Wu et al. 2019c) utilizes multi-head self-attention mechanisms for news and user modeling. Moreover, certain models introduce more complex architectures to capture news and user representations. For instance, HieRec (Qi et al. 2021) adopts a three-layer hierarchical framework that integrates topics and subtopics to enhance user modeling. MINER (Li et al. 2022b) employs a poly-attention mechanism with disagreement regularization and category-aware attention weighting to generate 32 distinct user interest vectors. Despite the ability of deep learning-based news recommendation methods to automatically learn news representations, they do not fully account for the higher-order relationships between user and news interactions. Therefore, some researchers endeavor to construct graph data structures to uncover deeper user interest characteristics.

In news recommendations, graph structures are typically employed to delineate the interactive relationships among multiple users and news items. The high-order connectivity between users and news encapsulates many features and n contextual information within such graphs. Graph neural networks (GNNs) have ascended as potent instruments for encoding graph data, facilitating the extraction of intricate features from interconnected nodes. In recent years, the prowess of GNNs in processing graph structures has garnered significant attention due to their robust feature extraction capabilities, which are predicated on both node attributes and the overarching graph architecture (Ji et al. 2021; Ge et al. 2020a; Hu et al. 2020b, a; Li et al. 2022a, 2021; Wu et al. 2023b). For instance, GERL (Ge et al. 2020b) not only derives text-based representations from news titles and topics but also capitalizes on a user-news graph to generate graph-based representations. Similarly, GNewsRec (Wu et al. 2021b) constructs a heterogeneous graph that interlinks users, news, and topics to unearth long-term user interests. User-as-graph (Wu et al. 2021b) innovates by introducing a heterogeneous graph pooling method to amalgamate the local news graph derived from a user's reading history. Additionally, DIGAT (Mao et al. 2022) enriches candidate news content by utilizing a semantic augmentation graph, which is constructed using methods that capture semantic relatedness. GLORY (Yang et al. 2023) represents a substantial leap forward in this domain by integrating global graph structures to augment news recommendations. GLoCIM (Yang et al. 2024) design a gated network to integrate long-chain interest with neighbor interest to achieve the collaborative interest among similar users.

Due to the powerful representation skills of GNNs in capturing the complex interactions between users and news in recommendation systems, this paper chooses to use GNN as the core model for personalized news suggestions.

2.2 News sentiment classification and recommendation (NSCR) systems

In recent years, sentiment analysis has become crucial for understanding and interacting with online content. Traditional news websites often do not incorporate sentiment analysis into their articles. In contrast, platforms like Reddit (Leavitt 2016) have recognized their significance, implementing upvotes and downvotes to gauge audience sentiment. The inclusion of sentiment factors in the recommendation system reflects that the platform has begun to pay attention to users' emotions for the recommended content.

Sentiment and sentiment analysis have been increasingly explored in recommendation systems to capture the affective aspects of user preferences and content. Sentiment analysis models, such as VADER and BERT-based classifiers, can identify the sentimental tone of the text, providing valuable insights into user preferences. SentiRec proposes a neural news recommendation method for the perception of sentiment diversity, which can recommend news with more diverse emotions (Wu et al. 2020a). Twittener (Tien et al. 2024), an improved news aggregator, improves the reading experience and time efficiency of users reading news online by implementing text-to-speech technology, sentiment analysis, and a hybrid recommendation system. However, these papers are still stuck in document-level and sentence-level sentiment analysis, lacking fine-grained sentiment analysis.

While sentiment and sentiment analysis have been integrated into various recommendation models, their application in graph-based frameworks remains limited. Existing studies have primarily focused on integrating document-level or sentence-level sentimental features into non-graph-based or graph-based methods (Tao and Alatas 2024b), leaving a gap in the literature regarding integrating fine-grained sentimental features into graph neural networks for recommendation tasks.

This paper addresses these gaps by proposing a novel approach to incorporating sentiment vectors into the graph-based framework. This integration enhances the framework's ability to deliver personalized news recommendations that consider textual and sentimental aspects of content. Potential benefits of such integration include a deeper understanding of user preferences and more accurate modeling of user-item interactions. By building on the strengths of existing methods and addressing their limitations, our proposed approach aims to provide a comprehensive and effective solution for

personalized news recommendations that leverage the rich sentimental context of news articles.

3 Methods

In this section, we describe our proposed approach for integrating sentiment vectors into a graph-based personalized news recommendation system. Our method involves several key steps: sentiment vector extraction, news and entity encoder, sentiment vector integration, graph-based news and entity encoder, user encoder, and a news recommendation part, as illustrated in Fig. 2. We detail each of these steps below.

3.1 Problem formulation

The click history of a user u is encoded as $\mathbf{H}_u = [d_1, d_2, \ldots, d_H]$, where H indicates total count of historical news articles. Each news item d_i contains three constituent elements: 1) a textual title represented as a token sequence $\mathbf{T}_i = [w_1, w_2, \ldots, w_T]$ with T word tokens, 2) an associated entity sequence $\mathbf{E}_i = [e_1, e_2, \ldots, e_E]$ comprising E entities, 3) a parallel entity sentiment vector $\mathbf{E}_i^{emo} = [e_1^{emo}, e_2^{emo}, \ldots, e_E^{emo}]$ capturing affective attributes for corresponding entities. The core computational objective involves estimating a relevance score $s_{u,c}$ that quantifies the predicted engagement likelihood between user u and given candidate news article d_c. This predictive framework essentially operates as a ranking system that orders multiple

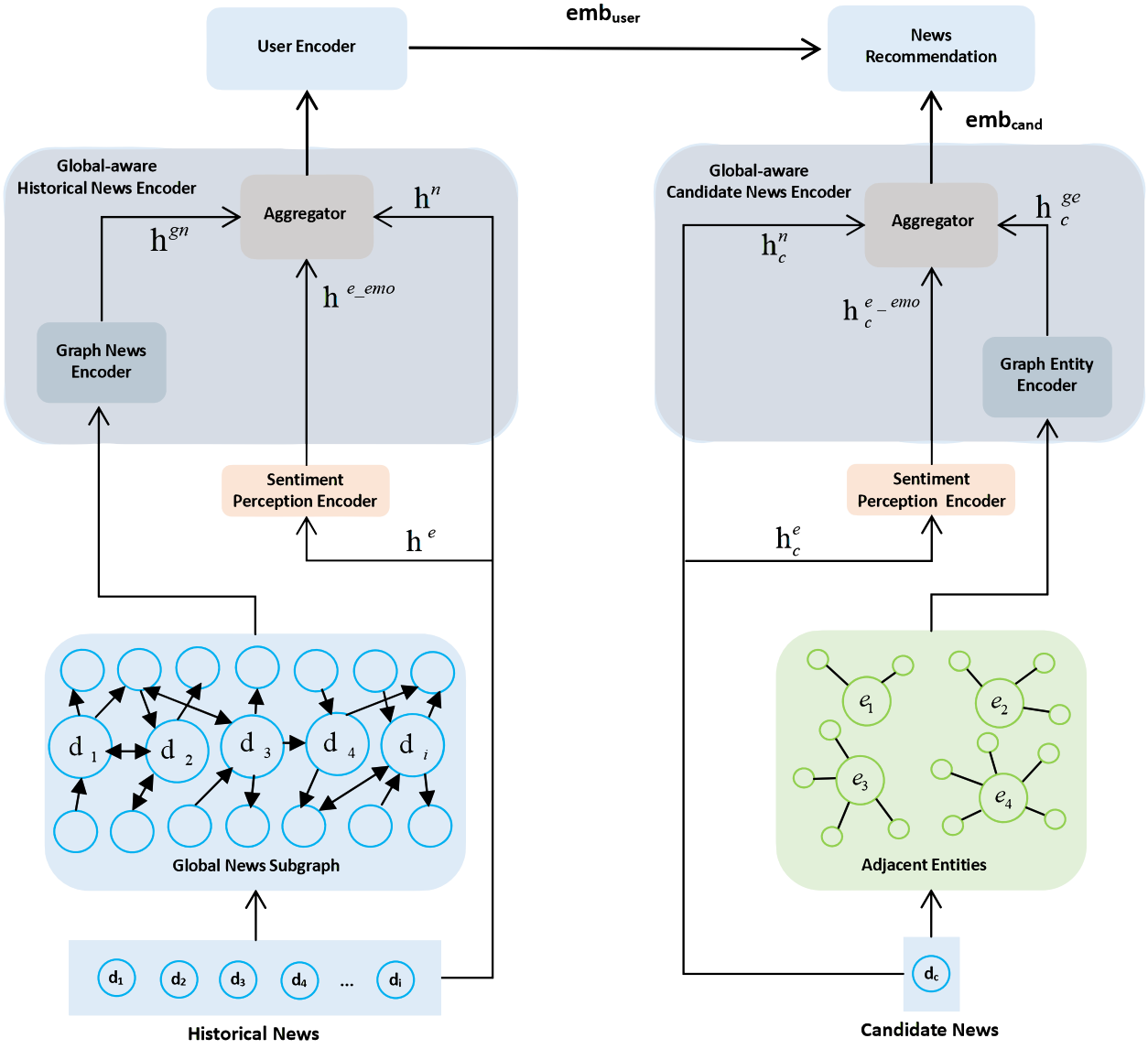


Fig. 2 A synopsis of the SentiEntityRec model

candidate news items through calculated click-through probability estimates to optimize recommendation effectiveness.

3.2 Overview of the SentiEntityRec framework

SentiEntityRec (Sentiment-aware Graph Neural News Recommendation) enhances traditional news recommendation systems by integrating sentiment vectors into a graph neural network (GNN) framework. The core idea is to enrich the entity representations used in GNN with sentimental information derived from a pre-trained sentiment analysis model. This approach allows the system to recommend news articles based on their relevance and sentimental resonance with the user, leading to a more engaging and personalized user experience.

The SentiEntityRec architecture consists of two main components:

1. Pre-trained Sentiment Analysis Model: This component analyzes the sentimental polarity of a news entity, generating a sentiment vector
2. Sentiment Perception Entity Encoding: The sentiment vector information is integrated into the entity encoding process, modifying the node representations in the GNN

3.3 Sentiment perception component(SPC)

In the process of entity-level emotion classification of the MIND dataset, we encounter the following problems: the absence of a title entity in the MIND dataset, and the existing entity-level emotion analysis cannot classify entity perception for a long text.

So, we designed a sentiment perception component to solve the above two problems. The phase of this algorithm consists of three main steps: 1) entity localization and extraction, 2) text segmentation, and 3) sentiment computing. Entity localization and extraction: Extract the entities in the news title first. If not, check the entities in the abstract and filter out those that appear in the header. Figure 3. shows the flowchart of SPC. Text segmentation: Sentence-level segmentation of excessively long titles or abstract texts. Sentiment computing: according to the text where the entity is located, the entity-level sentiment analysis is carried out based on NewsMTSC (Hamborg et al. 2021). An illustration of the algorithm is as follows:

Here's an example of how this part works, as shown in Table 1. For each news article, the sentiment perception component processes the news text to extract the sentiment vector of the recognized entities. The output is a probability distribution over the sentiment categories [negative, neutral, positive], which is then summarized into a set of sentiment-aware features representing the sentiment vector

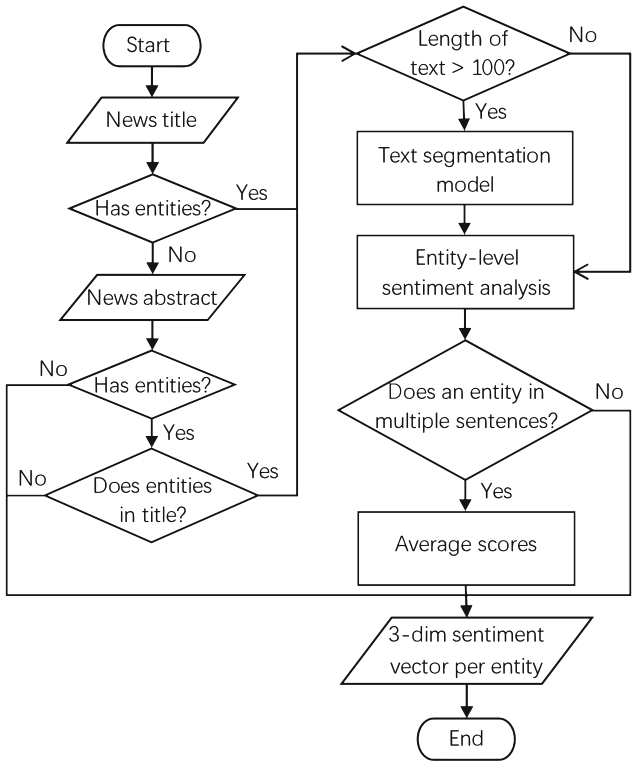


Fig. 3 The flowchart of SPC

of each entity. For example:

$$e_i^{\text{emo}} = [0.452, 0.3652, 0.1828]$$

Where 0.452, 0.3652, and 0.1828 represent the probabilities of negative, neutral, and positive emotions, respectively.

3.4 Sentiment perception and semantic entity encoding

First, we use sentiment perception component to extract entities $\mathbf{E}_i = [e_1, e_2, ..., e_E]$ and get the entity sentiment vectors $\mathbf{E}_i^{emo} = [e_1^{emo}, e_2^{emo}, ..., e_E^{emo}]$. An entity embedding layer is used to merge information from a knowledge graph. The entity sequence $\mathbf{E}_i = [e_1, e_2, ..., e_E]$ is converted to a semantic entity embedding vector $\mathbf{x}_e = [x_1^e, x_2^e, ..., x_E^e]$ sequence using pre-trained TransE (Bordes et al. 2013) entity embedding based on WikiData[5].

Secondly, we integrate entity sentiment vectors into the entity embedding vectors (Dang et al. 2021). Based on the entity sentiment vectors $\mathbf{E}_i^{emo}$ and the semantic entity embedding vectors $\mathbf{x}_e$, the sentiment-aware entity encoding $\mathbf{x}_{e_emo}$ is computed as follows:

$$\mathbf{x}_{e_emo} = \text{concat}(\mathbf{x}_e; \mathbf{E}_i^{emo}) \quad (1)$$

[5] https://www.wikidata.org/wiki/Wikidata:MainPage

Algorithm 1 Entity Sentiment Vector Extraction

Require: news_data: News, sen_model: Sentiment analysis model, seg_model: Long text segmentation model, max_len: Maximum text length threshold
Ensure: entity_sentiments
$max_len \leftarrow 100$
$entity_sentiments \leftarrow \{\}$
for each $item$ in $news_data$ **do**
▷ Entity localization and extraction
  $entities \leftarrow$ parse_json($item$["$title_entities$"]) $\vee$ $\{e \mid e \in$ parse_json($item$["$abstract_entities$"]) $\wedge e \in item$["$title$"]$\}$
  if $item$["$title_entities$"] $\neq \emptyset$ **then**
    $entities \leftarrow item$["$title_entities$"]
    $text \leftarrow item$["$title$"]
  else if $item$["$abstract_entities$"] $\cap\, item$["$title$"] $\neq \emptyset$ **then**
    $entities \leftarrow item$["$abstract_entities$"] $\cap\, item$["$title$"]
    $text \leftarrow item$["$title$"]
  else
    $entities \leftarrow item$["$abstract_entities$"]
    $text \leftarrow item$["$abstract$"]
  end if
▷ Text segmentation
  if len($text$) $\leq max_len$ **then**
    $sentences \leftarrow [text]$
  else
    $sentences \leftarrow$ seg_model.segment($text$)
  end if
▷ Sentiment computing
  for each $entity$ in $entities$ **do**
    $pols, vecs \leftarrow [], []$
    for each $sentence$ in $sentences$ **do**
      if $entity \in sentence$ **then**
        $p, v \leftarrow$ sen_model.analyze($entity, sentence$)
        pols.append(p); vecs.append(v)
      end if
    end for
    if $vecs \neq \emptyset$ **then**
      $avg_vec \leftarrow \sum(vecs)/\text{len}(vecs)$
    else
      $avg_vec \leftarrow [0.0, 0.0, 0.0]$
    end if
    if $pols \neq \emptyset$ **then**
      $dominant \leftarrow \arg\max_{p \in \text{set}(pols)} \text{count}(p)$
    else
      $dominant \leftarrow$ "None"
    end if
    entity_sentiments[$entity$] $\leftarrow (dominant, avg_vec)$
  end for
end for

Table 1 An example of entities and entity sentiment polarities extracted from a news headline

News Headline	Recognized Entity	Sentiment Polarity	Sentiment Probability
Alibaba reports strong Q2 growth while Tencent faces regulatory challenges	Alibaba	Positive	[0.0239, 0.4144, 0.5617]
	Tencent	Negative	[0.5553, 0.3328, 0.1119]

where concat() denotes the concatenation operation, and $\mathbf{x}_{e_emo}$ are the fused features.

Third, we use an entity multi-head self-attention network to learn sentiment-aware entity representations $\mathbf{X}_{e_emo}$. Self-Attention Mechanism:

$$Q = W_q \mathbf{x}_{e_emo}, \quad K = W_k \mathbf{x}_{e_emo}, \quad V = W_v \mathbf{x}_{e_emo} \tag{2}$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{3}$$

Where W_q, W_k, W_v are the projection matrices for query, key, and value, d_k is the dimension of the key, and softmax is the normalization function. Multi-Head Attention:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O \tag{4}$$

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \tag{5}$$

Where h is the number of attention heads, W_i^Q, W_i^K, W_i^V are the projection matrices for the i-th attention head, and W^O is the output projection matrix. Final Output:

$$O = \text{LayerNorm}(F + \text{Dropout}(\text{MultiHead}(Q, K, V))) \tag{6}$$

$$\text{X}_{e_emo} = \text{LayerNorm}(O + \text{Dropout}(\text{FFN}(O))) \tag{7}$$

Where LayerNorm is layer normalization, Dropout is dropout regularization, and FFN is the feed-forward neural network.

Lastly, obtain the news entity representation $\mathbf{h}^{e_emo}$ by aggregating several entities using an attention network:

$$\mathbf{h}^{e_emo} = \sum_{i=1}^{E} \alpha_i^e \mathbf{x}_i^e, \quad \alpha_i^e = \frac{\exp(\mathbf{q}_e^\top \tanh(\mathbf{W}^e \cdot \mathbf{X}_{e_emo}^i))}{\sum_{j=1}^{E} \exp(\mathbf{q}_e^\top \tanh(\mathbf{W}^e \cdot \mathbf{X}_{e_emo}^j))} \tag{8}$$

where the query vector is $\mathbf{q}_e$ and the trainable matrix is $\mathbf{W}^e$. Furthermore, $\mathbf{X}_{e_emo}^i$ represents the i-th entity representation of $\mathbf{X}_{e_emo}$.

This sentiment perception encoding process ensures that the entity representations used in the GNN capture the semantic relationships between entities and reflect their sentimental tone. It makes the recommendation process more sensitive to the user's sentimental context.

The semantic entity encoding $\mathbf{h}^e$ is obtained similarly to the sentiment-aware entity encoding. The schematic diagram of the semantic entity encoder and sentiment perception entity encoder is shown in Fig. 4.

3.5 News encoding

First, use a pre-trained GloVe (Pennington et al. 2014) word embeddings as initialization to convert a news text from a sequence of words $\mathbf{T}_i = [w_1, w_2, \ldots, w_T]$ into a sequence of word embedding vectors $\mathbf{x}_w = [x_1^w, x_2^w, \ldots, x_T^w]$.

Next, learn word representations $\mathbf{X}_w$ using a text multi-head self-attention (MSA) (Vaswani et al. 2017) layer, as in Eq.9.

Finally, aggregate news word representations using an attention network to obtain news representation $\mathbf{h}^n$, as in Eq.10. The news encoder is demonstrated in Fig. 4.

3.6 Graph neural network

An important component of SentiEntityRec is integrating sentiment vectors into entity encoding.

3.6.1 News graph Encoder

To summarize the reading histories of users, we need to build a news graph. We use $G_n = (V_n, E_n)$ to represent this news graph, in which V_n and E_n stand for the news article and edge sets, respectively. We use a directed edge (v_i, v_j) to represent that the same user read news v_j immediately after reading news v_i. The weight of the edge is determined by how often this occurs in all read histories.

The news graph is derived from the reading history of each user. Due to the large number of news articles, we use a subgraph $G_{sub}^u = (V_{sub}^u, E_{sub}^u)$ to represent a user's reading history. From the global news graph, we select news neighbors of multiple hops for each news item in $\mathbf{H_u}$ of user u. Based on the weight of the edge, we choose the best M_n neighbors for every hop. The subgraph is then encoded using graph neural networks (GNN) (Li et al. 2015a) to produce a global news embedding $\mathbf{h}^{gn}$.

$$\mathbf{h}^{gn} = \text{GNN}(\mathbf{h}^n) \tag{9}$$

In the context of the graph encoder, we employ a gated recurrent unit (GRU) (Cho et al. 2014) from the gated graph neural network (GGNN) (Li et al. 2015b) to acquire sequence-oriented hidden behavior details.

$$\mathbf{h}_i^{(0)} = \mathbf{h}_i^n \tag{10}$$

$$\mathbf{m}_i^{(l+1)} = \sum_{j \in N(i)} W_g \cdot \mathbf{h}_j^{(l)} \tag{11}$$

$$\mathbf{h}_i^{(l+1)} = \text{GRU}(\mathbf{m}_i^{(l+1)}, \mathbf{h}_i^{(l)}) \tag{12}$$

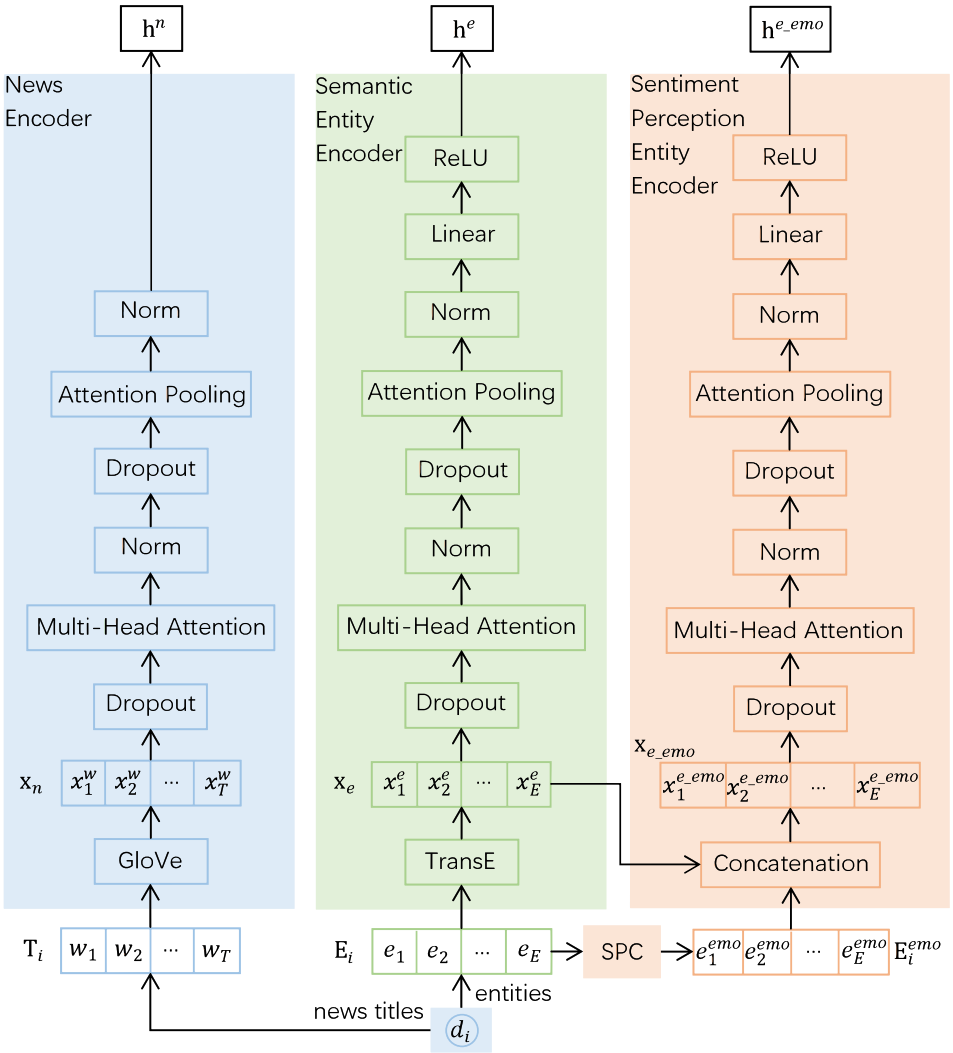


Fig. 4 A schematic diagram of encoders of Semantic Entity, Sentiment Perception Entity, and News

$\mathbf{h}^{(0)}$ is initialized with $\mathbf{h}^n$. $N(i)$ is the set of nodes that neighbor node i in subgraph G^u_{sub}. l is the layer number, and W_g is a trainable matrix.

3.6.2 Semantic entity graph encoder

A newly generated news article often lacks user interaction history, but usually has entities. The relationship between candidate news entities and other news entities is judged by comparing the relationship between candidate news and other news. So, we built a semantic entity graph similar to the news graph.

We took an undirected graph and applied a similar method to create an entity graph as we did with the news graph. V_e and E_e are the sets of entities and edges that make up the entity graph, which is represented as $G_e = (V_e, E_e)$. Entity edges are generated based on each user's reading history on the training dataset. An undirected edge (v_i, v_j) indicates that the entity of the previous news article is v_i, and the entity of the one that follows is v_j. The number of news edge occurrences determines the edge weight of (v_i, v_j).

Here is a simple example: Figure 5 is a simple case of a news graph, a semantic entity graph, and sentiment perception news entities. The reading sequences $[n_2, n_1, n_3, n_4]$ and $[n_7, n_6, n_1, n_5]$ from users U8125 and U8312, respectively, are used by global news graph. Semantic global entity graph shows six entities from news n_1, n_2, n_3, and displays the entities $[e_1]$ and $[e_2, e_3]$ from news n_1 and n_2, respectively.

In the global entity graph, the average degree of nodes is much higher because there are comparatively fewer entities than news articles. Therefore, it is more practical and concise to use adjacent entities directly rather than entity subgraphs. Analogously, we employ the same entity embedding layer as the entity encoder, choosing the best M_e neighbor entities

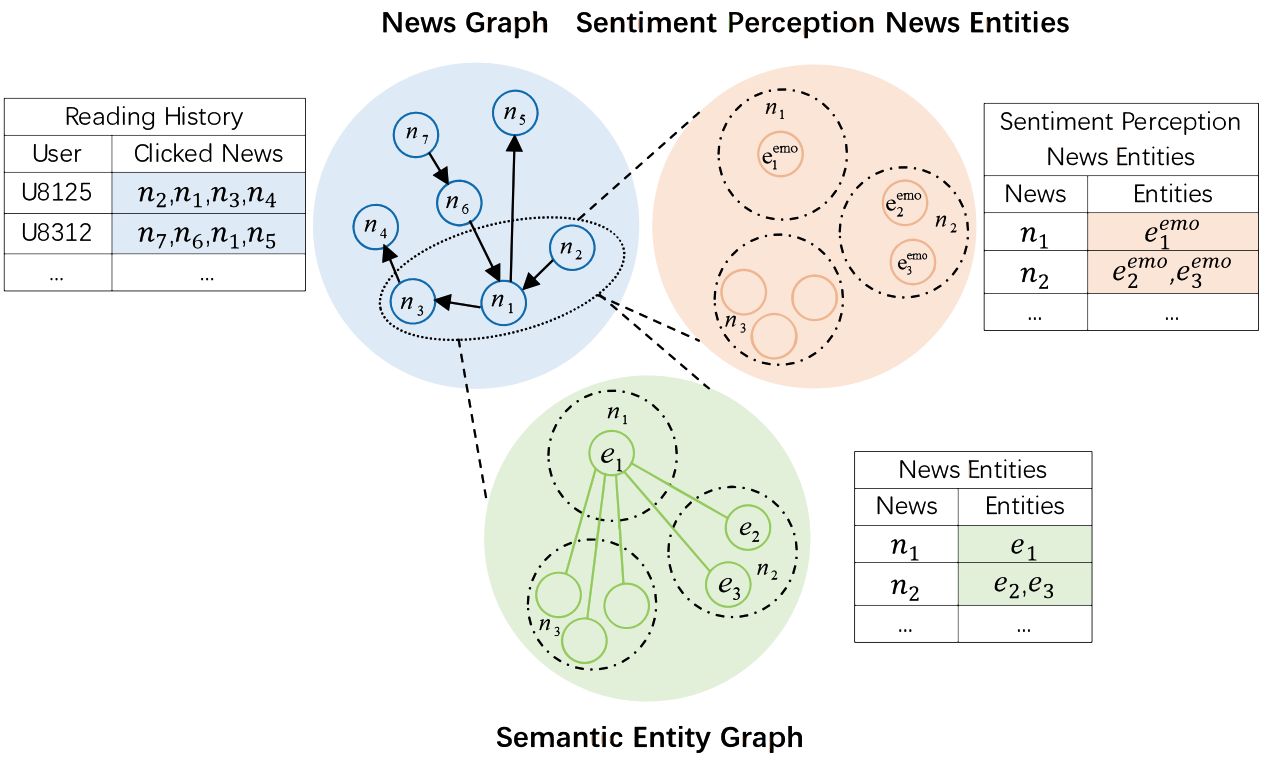


Fig. 5 A case of News Graph, Semantic Entity Graph, and Sentiment Perception News Entities

for every entity in candidate news d_c based on edge weights. We can get the entity embedding $\mathbf{x}_e = [x_1^e, x_2^e, \ldots, x_{E*M_e}^e]$. Then, we use an MSA to learn the entity token representations $\mathbf{X}_e$, as in Eq.9, and an attention network to aggregate the neighboring entity representations $\mathbf{h}^{ge}$, as in Eq.10.

3.6.3 Historical news aggregator

We have the news representation $\mathbf{h}^n$, the sentiment-aware entity representation $\mathbf{h}^{e_emo}$, and the semantic news graph representation $\mathbf{h}^{gn}$ for each historical news article d_k. We then aggregate these three representations $\mathbf{h}^k = [\mathbf{h}^n, \mathbf{h}^{e_emo}, \mathbf{h}^{gn}]$ to learn the historical news representation $\mathbf{n}_k$ from a global perspective using an attention pooling network:

$$\mathbf{n}_k = \sum_{i=1}^{4} \alpha_i^h \mathbf{h}_i^k, \quad \alpha_i^h = \frac{\exp(q_h^\top \tanh(W^h \cdot \mathbf{h}_i^k))}{\sum_{j=1}^{4} \exp(q_h^\top \tanh(W^h \cdot \mathbf{h}_j^k))} \tag{13}$$

where q_h is the query vector and W^h is a trainable matrix. Additionally, the i-th concatenated representation of $\mathbf{h}^k$ is indicated by $\mathbf{h}_i^k$.

3.6.4 Candidate news aggregator

There is the news representation $\mathbf{h}_c^n$, the sentiment-aware entity representation $\mathbf{h}_c^{e_emo}$, and semantic entity graph representation $\mathbf{h}_c^{ge}$ for every candidate news article d_c. $\mathbf{h}_c^{ge}$ is added to the candidate news representation to obtain the semantic correlation between the candidate news entities and the news reading history of a user. Then, we aggregate these three representations $\mathbf{h}^c = [\mathbf{h}_c^n, \mathbf{h}_c^{e_emo}, \mathbf{h}_c^{ge}]$, using an attention pooling network to learn the candidate news representation $\mathbf{emb}_{cand}$, as in Eq.15. A synopsis flowchart from candidate news d_c to candidate news representation $\mathbf{emb}_{cand}$ is exhibited in Fig. 6.

3.6.5 User encoder

We then briefly describe the user encoder architecture in SentiEntityRec. By Wu et al. (2019c), we obtain news embedding $\mathbf{n}_k$ of each news in a given user reading history $\mathbf{H}_u$, and then, similar to Eq.10, we use a multi-head attention layer and an attention pooling layer to eventually learn the user representation $\mathbf{emb}_{user}$. A synopsis flowchart from user u to user representation $\mathbf{emb}_{user}$ is displayed in Fig. 7.

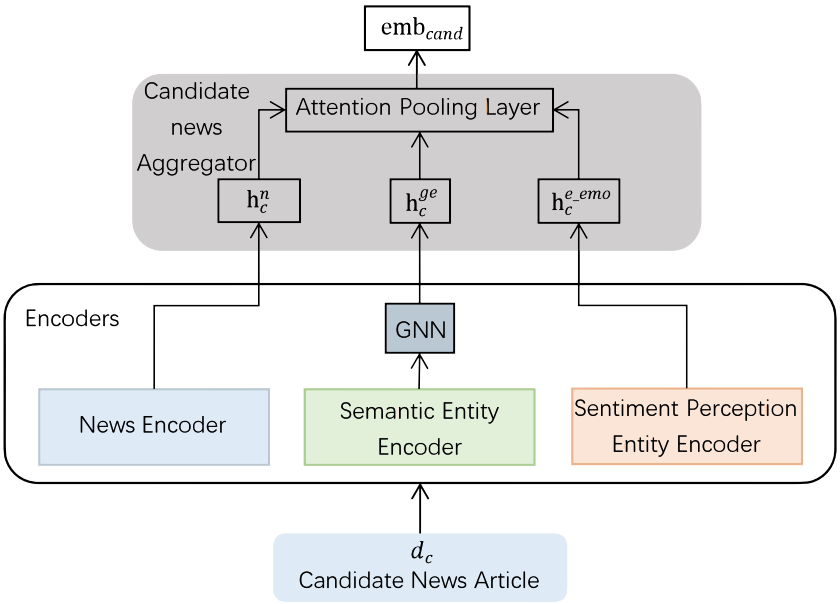


Fig. 6 A synopsis flowchart from candidate news d_c to candidate news representation $\mathbf{emb}_{cand}$

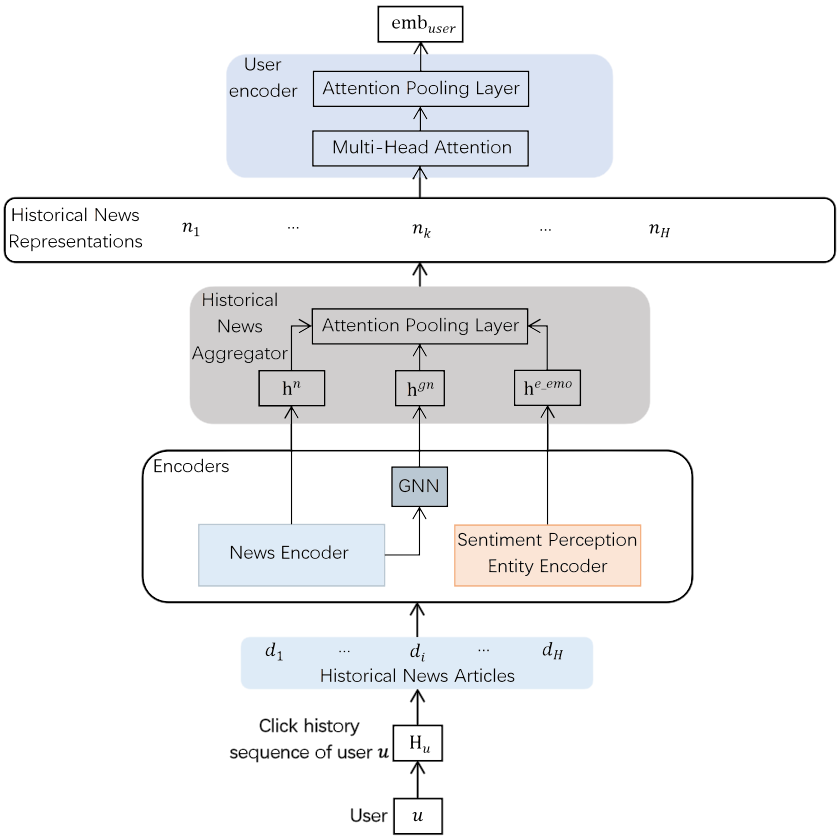


Fig. 7 A synopsis flowchart from user u to user representation $\mathbf{emb}_{user}$

3.7 Click prediction and negative log likelihood loss

The SentiEntityRec model employs a Noise Contrastive Estimation (NCE) loss function for training. This approach is particularly suitable for news recommendation tasks. Because it can effectively handle a large number of candidate news items by sampling negative examples while maintaining the model's ability to learn from positive user-news interactions. Consistent with earlier studies, we employ the negative sampling technique in our training procedure (Li et al. 2015b). K_{neg} non-clicked candidate news within the same session are randomly chosen as negative samples $[n_i^1, n_i^2, \ldots, n_i^{K_{neg}}]$, at the same time each clicked candidate news is designated as a positive sample n_i^+ for each behavioral session in the training dataset. Then, using an inner product between user embedding $\mathbf{emb}_{user}$ and all samples $\mathbf{emb}_{cand}$, which are made up of one positive sample and K_{neg} negative samples, we estimate the click score for each news article using the formula $\hat{y}_i = [\hat{y}_i^+, \hat{y}_i^1, \hat{y}_i^2, \ldots, \hat{y}_i^{K_{neg}}]$. In the end, we optimize the positive samples log-likelihood loss $\mathcal{L}_{NCE}$ across the training dataset during model training:

$$\hat{y}_i = \text{softmax}(\text{DotProduct}(\text{emb}_{user}, \text{emb}_{cand})), \tag{14}$$

$$\mathcal{L}_{NCE} = -\sum_{i=1}^{N} \log \frac{\exp(\hat{y}_i^+)}{\exp(\hat{y}_i^+) + \sum_{j=1}^{K_{neg}} \exp(\hat{y}_i^j)}. \tag{15}$$

Using the trained SentiEntityRec, calculate the probability of click scores of a user clicking on candidate news. Generate personalized recommendations by ranking news articles based on their relevance to the user, with higher importance given to articles that align with the user's interests and sentimental state.

4 Experiments

4.1 Datasets

To evaluate the effectiveness of the SentiEntityRec framework, we conducted experiments using a widely recognized benchmark real-world dataset for news recommendation: MIND (Microsoft News Dataset) (Wu et al. 2020b).

MIND Dataset[6]: MIND is a large-scale dataset collected from the Microsoft News[2] platform. MINDs come in two varieties. The first is MIND-large, a full version of MIND that includes a million sampled users who clicked on at least five news items in the six weeks between October 12 and November 22, 2019. The other is MIND-small, a scaled-down version of MIND that includes 50,000 sampling users and their behavior logs.

The key statistics of the dataset are displayed in Table 2.

4.2 Evaluation metrics

According to the setting from Wu et al. (2020b), we used several standard evaluation metrics commonly employed in recommendation systems to measure the performance of SentiEntityRec: Mean Reciprocal Rank (MRR), normalized discounted cumulative gain (nDCG@N) with $N = 5, 10$, and area under the ROC curve (AUC).

4.3 Baselines

To demonstrate the effectiveness of SentiEntityRec, we compared its performance with three kinds of baseline methods:

Feature-based Methods: (1) LibFM (Rendle 2012), a well-known recommendation system that predicts user preferences for news articles based on their features by using factorization machines; (2) DeepFM (Guo et al. 2017), a recommendation system based on neural networks that combines factorization machines and deep learning to improve the accuracy of news article recommendations.

Neural Recommendation Methods: (1) GRU (Okura et al. 2017), learning user representations from a sequence of news items clicked on using a GRU network; (2) DKN (Wang et al. 2018), utilizing a CNN with knowledge awareness to obtain news representation from knowledge entities and news text; (3) NPA (Wu et al. 2019b), learning news and user representation based on a personalized attention network; (4) NAML (Wu et al. 2019a), learning user and news representation based on various news article features utilizing the multi-view attention network; (5) LSTUR (An et al. 2019), employing a GRU network to model users' long- and short-term interests simultaneously; (6) NRMS (Wu et al. 2019c), learning user and news representation using a multi-head self-attention network; (7) FIM (Wang et al. 2020), employing stacked dilated convolutions to perform fine-grained interest using hierarchical representations of news articles; (8) HieRec (Qi et al. 2021): modeling each

Table 2 Dataset statistics

	MIND-small	MIND-large
News	65,238	161,013
Users	50,000	1,000,000
Clicks	347,727	24,155,470
Entities	1,336,942	3,299,687
News Information	title, category, abstract, body	

[6] https://msnews.github.io/

user's diverse and multi-grained interests based on a three-level hierarchical structure; (9) MINER w/o PLM (Li et al. 2022b): modeling multiple user interests based on a poly attention scheme with disagreement regularization. For a fair comparison, we choose the version devoid of the pre-trained language model (PLM).

Graph-based Methods: (1) GERL (Ge et al. 2020a), improving user and news representations by combining a graph attention network with transformer architecture and a bipartite graph; (2) GNewsRec (Hu et al. 2020a), utilizing a heterogeneous graph to model user-news-topic interactions and learning high-order user and news representations with graph neural networks; (3) User-as-Graph (Wu et al. 2021b), utilizing a customized heterogeneous graph built from user behavior and a novel heterogeneous graph pooling technique; (4) DIGAT (Mao et al. 2022), employing a dual-graph interaction process and news- and user-graph channels to effectively match news-user representations;(5) GLORY (Yang et al. 2023) enhances recommendation accuracy by integrating global user interaction patterns-captured through a global news graph and a global entity graph-with local textual and entity representations via gated graph neural networks and multi-head self-attention network;(6) GLoCIM (Yang et al. 2024) combines neighbor interest with long chain interest distilled from a global click graph, leveraging the collaboration among similar users to enhance news recommendation. Similarly, we take the same global view to build graph neural networks in this paper.

These baselines represent a range of popular and cutting-edge methods in news recommendation, providing a comprehensive comparison against SentiEntityRec.

4.4 Experimental environments

SentiEntityRec was implemented using the PyTorch framework, with the GNN components built using PyTorch Geometric. The pre-trained NewsMTSC model was utilized to extract sentiment vectors from entities in news titles, which were then integrated into the entity encodings. The sentence length threshold using the sentence segmentation models was set to 100. Following previous work (Yang et al. 2023) and our experiment, we consider a user's most recent 50 clicked news articles as their reading history ($L_{his} = 50$) and limit each news title to a maximum of 30 tokens ($L_{title} = 30$). For each news article, up to 5 entities are utilized ($L_{entity} = 5$), and in constructing the news subgraph, we set the number of neighboring news to 8 ($M_n = 8$) and the number of hops to 2 ($K = 2$), with 10 neighboring entities ($M_e = 10$).

Our GNN architecture employs a three-layer GGNN. To initialize on the MIND dataset, we employ 100-dimensional pre-trained TransE entity embeddings and 300-dimensional pre-trained GloVe word embeddings. The model is trained using the Adam optimizer with an initial learning rate of $2e^{-4}$, which is linearly decayed following a 10% warm-up phase, and a negative sampling rate of 4 ($K_{neg} = 4$). All hyperparameters were selected based on validation set performance.

4.5 Overall Performance

Table 3 presents the overall performance of SentiEntityRec and the baseline models on MIND. The percentage symbol (%) is not used when expressing the tabulated values. The best overall results are bolded, and the best baseline results are underlined.

Table 3 summarizes the overall performance of SentiEntityRec compared to baseline models on the MIND datasets. First, our experiments confirm that deep learning-based recommendation models significantly outperform traditional feature-based methods by automatically learning effective representations from complex news content. Second, while previous models-such as GLORY-leverage global graph structures to capture hidden relationships between users and news, SentiEntityRec further enhances these representations by integrating an entity-level sentiment vector. This additional sentiment context improves the alignment between user interests and recommended content, as evidenced by higher AUC and nDCG scores. The results indicate that the integration of sentiment vectors into entity encodings leads to improvements in recommendation accuracy and ranking performance. The results also show that the sentiment-aware entity encodings enable the model to capture both semantic relevance and sentiment resonance of news articles, resulting in more personalized and engaging recommendations.

In summary, the step-by-step improvements from manually created features to deep learning feature extraction, global graph integration, and sentiment information inclusion demonstrate SentiEntityRec's superior performance in personalized news recommendation.

4.6 Ablation study

To assess the impact of the sentiment vector integration, we conducted an ablation study by comparing the performance of SentiEntityRec (**full model**) with its variant. The variant: **w/o local** e^{emo}, means the entity representation without sentiment perception.

This variant helps measure how much the sentiment perception component improves recommendation accuracy. As seen from the data in Table 4, the local entity with sentiment perception plays a promoting role. This demonstrates the effectiveness of sentimental information in enhancing news recommendations' performance.

Table 3 Evaluate performance at all baselines in MIND. Because we reuse the results of existing work, some values will be lost: * is taken from Li et al. (2022b); † is taken from Wu et al. (2021b); ‡ is taken from Mao et al. (2022). Underline means the best performance of baseline methods, while **bold** means the best in all metrics

Method	*MIND-small*				*MIND-large*			
	AUC	MRR	nDCG@5	nDCG@10	AUC	MRR	nDCG@5	nDCG@10
LibFM	59.74	26.33	27.95	34.29	61.85	29.45	31.45	37.13
DeepFM	59.89	26.21	27.74	34.06	61.87	29.30	31.35	37.05
GRU	61.51	27.46	30.11	36.61	65.42	31.24	33.76	39.47
DKN	62.90	28.37	30.99	37.41	64.07	30.42	32.92	38.66
NPA	64.65	30.01	33.14	39.47	65.92	32.07	34.72	40.37
NAML	66.12	31.53	34.88	41.09	66.46	32.75	35.66	41.40
LSTUR	65.87	30.78	33.95	40.15	67.08	32.36	35.15	40.93
NRMS	65.63	30.96	34.13	40.52	67.66	33.25	36.28	41.98
FIM	65.34	30.64	33.61	40.16	67.87	33.46	36.53	42.21
HieRec	67.83	32.78	36.31	42.49	69.03	33.89	37.08	43.01
MINER w/o PLM*	68.07	32.93	-	42.62	-	-	-	-
GERL	65.27	30.10	32.93	39.48	68.10	33.41	36.34	42.03
GNewsRec	65.54	30.27	33.29	39.80	68.15	33.45	36.43	42.10
User-as-Graph†	-	-	-	-	69.23	34.14	37.21	43.04
DIGAT w/o PLM‡	67.82	32.65	36.25	42.49	-	-	-	-
GLORY	67.60	32.45	35.75	42.09	69.04	33.83	37.53	43.49
GLoCIM	67.74	32.50	36.01	42.32	69.06	34.13	37.55	43.58
SentiEntityRec	**68.23**	**32.95**	**36.61**	**42.63**	**69.46**	**34.22**	**37.93**	**43.98**

4.7 Analysis on entity and sentiment distribution

To better understand the role of entities in sentiment perception news recommendation, we analyzed the number of entities identified in news titles from MIND and their sentiment polarity distributions. Figure 8 presents the distribution of entity counts per news article, showing that the majority of news articles contain between 0 and 4 named entities, with a long-tail distribution indicating a small proportion of articles containing more than 9 entities.

Furthermore, Figure 9 illustrates the sentiment polarity distribution of the identified entities. We observed that neutral entities dominate in MIND datasets, followed by a small proportion of entities associated with positive and negative sentiments. These positive and negative subsets of sentiment entities differentiate our model from the state-of-the-art previous models. This imbalance in sentiment distribution suggests that real-world news articles tend to focus more on neutral narratives. The proportion distribution of entities with three sentiment polarities shows that it is necessary to distinguish entities in sentiment polarity.

These findings highlight the importance of incorporating entity-level sentiment analysis in news recommendation systems. Because the skewed distribution of sentiment-laden entities can affect the learning process of sentiment perception models, which influences user engagement and recommendation accuracy.

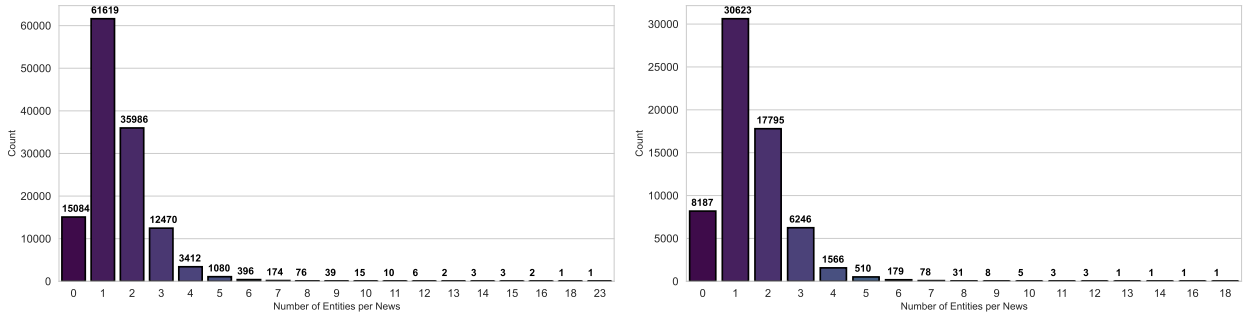
Table 4 The influence of SentiEntityRec sentiment component on MIND-small

Model Variant	AUC	MRR	nDCG@5	nDCG@10
SentiEntityRec	68.23	32.95	36.61	42.63
w/o e^{emo}	67.60	32.45	35.75	42.09

4.8 Hyperparameter analysis

In the experiment, the effect of neural networks and contrastive learning in our model is primarily influenced by three key hyperparameters: they are the number of news neighbors M_n, the number of hops K, and the number of adjacent entities M_e.

The impacts of two parameters M_n and M_e on the model's performance on the MIND dataset are illustrated in Figs. 10 and 11. We found that when $K = 2$ and $M_e = 10$ are fixed, SentiEntityRec achieves optimal performance at $M_n = 8$; Increasing the number of news neighbors further does not



(a) Entity Count Distribution in MIND-large

(b) Entity Count Distribution in MIND-small

Fig. 8 Distribution of entities per news identified from MIND

yield additional benefits, likely because too many neighbors introduce weakly related news that dilutes the effective associations in the subgraph. Similarly, with $K = 2$ and $M_n = 8$ fixed, the best performance is obtained when $M_e = 10$. This suggests that while adjacent entities help capture more relevant information, an excessive number of them adds noise that harms the model's performance.

Figure 12 illustrates how hyperparameter K impacts the performance of SentiEntityRec. Our experiments with various values of K indicate that $K = 2$ is optimal; Overly shallow subgraphs fail to capture the complete reading trajectory, whereas overly deep subgraphs incorporate too much noise and incur unnecessary computational costs.

4.9 Impact of graph encoder

We used two popular graph models, Graph Convolutional Network(GCN) (Kipf and Welling 2016) and Graph Attention Network(GAT) (Veličković et al. 2017), to model graphs compared with gated graph neural network (GGNN) (Li et al. 2015a) in SentiEntityRec.

Table 5 shows that GGNN outperforms the other graph models because it can learn the hidden sequential information from click behavior in users' reading histories. We also observe that as hops increase, training time and memory consumption grow exponentially, while performance declines. This indicates that the more hops, the better the model's performance.

5 Conclusion and future work

This study introduces SentiEntityRec, a novel sentiment perception graph neural news recommendation model. SentiEntityRec can not only capture behavioral and semantic associations but also consider the user's sentiment orientation toward a particular entity. This model realizes news recommendations' fine-grained emotion perception ability

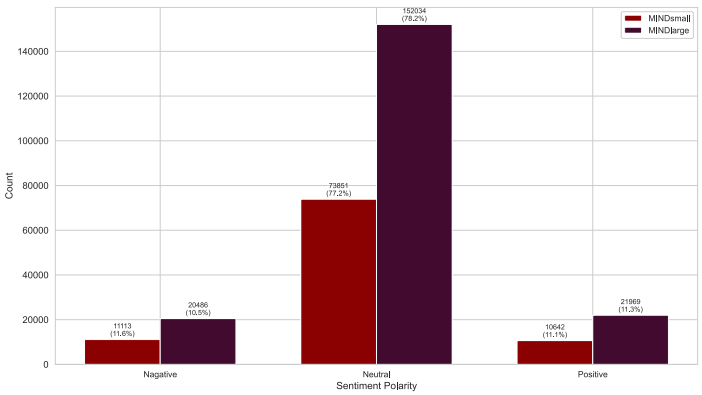
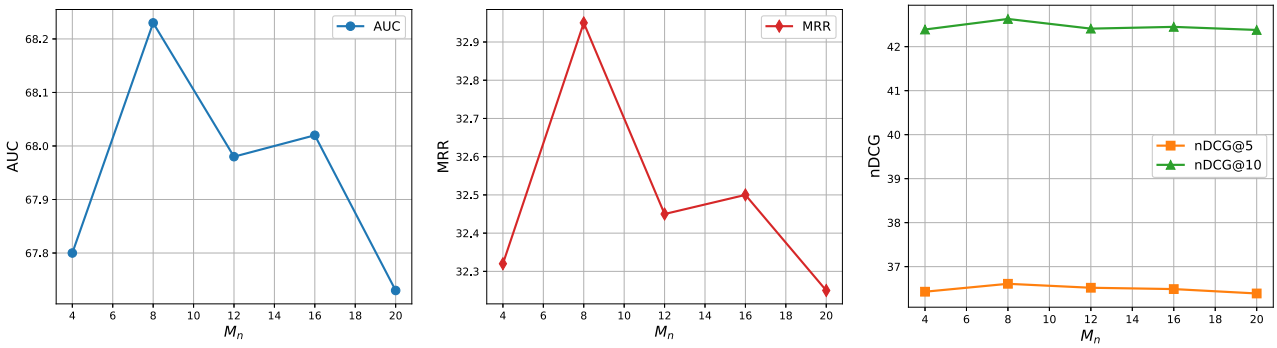
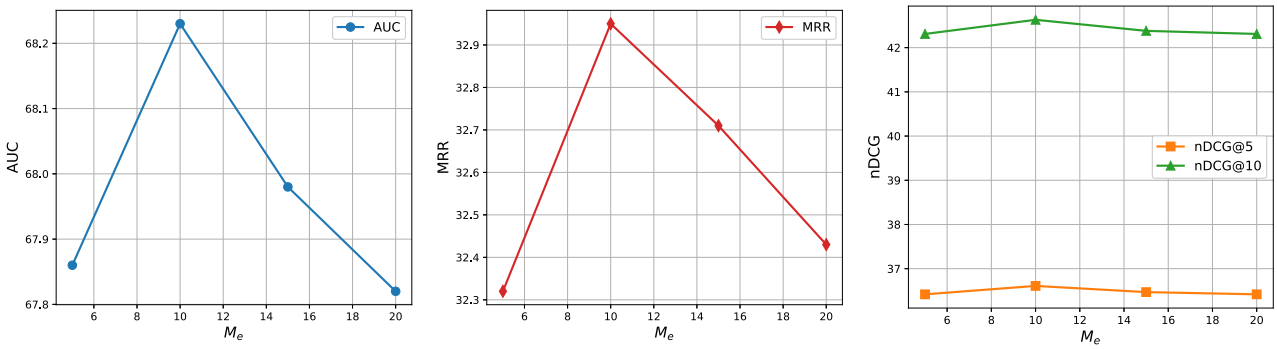


Fig. 9 Sentiment Polarity Distribution in MIND



(a) Different news neighbors on AUC (b) Different news neighbors on MRR (c) Different news neighbors on nDCG

Fig. 10 Impact of news neighbors



(a) Different adjacent entities on AUC (b) Different adjacent entities on MRR (c) Different adjacent entities on nDCG

Fig. 11 Impact of adjacent entities



(a) Different hops on AUC (b) Different hops on MRR (c) Different hops on nDCG

Fig. 12 Impact of hops

Table 5 Effects of Different Graph Construction Model performance comparison. The average time cost of training 1000 samples. Underline means the best performance of other models, while **bold** means the best in AUC

Model	Hops	AUC	Training time(s)	GPU Memory consumption(GB)
GCN	1	67.26	9.94	12.48
	2	67.42	21.01	17.43
	3	67.33	110.09	20.57
GAT	1	67.98	10.85	7.71
	2	68.09	21.78	14.50
	3	68.03	113.91	19.02
GGNN	1	68.01	11.01	7.01
	2	**68.23**	21.89	14.03
	3	68.11	126.38	21.16

at the entity level by enabling entities to have contextual sentiment perception ability and building a global sentiment perception entity graph. We perform tests on a public dataset to evaluate the effectiveness of our model. The experimental results prove that entities with contextual sentiment perception ability can improve the performance of the recommendation system. Although SentiEntityRec demonstrates improvements in key performance metrics, it requires relatively large computational resources, which will also limit the use of this model in a real-time news recommendation system. Besides, synonyms and related words between news entities cannot be well identified. Future work will focus on optimizing GNN scalability for real-time news recommendations and increasing the ability to recognize synonyms and related words.

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Author Contributions The first author is responsible for data collection and preprocessing, data verification and processing, model building, coding of the algorithms, experiments, the overall training and testing process, and writing the 1st draft of the paper; The second author assists methodology design and data evaluation; The third author is responsible for the overall project promotion, paper review and revision, model building and adjustment, protocol perform.

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Data Availability Data can be obtained by contacting corresponding author.

Declarations

Conflicts of Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Aggarwal CC et al (2016) Recommender systems, vol 1. Springer, Berlin

An M, Wu F, Wu C, Zhang K, Liu Z, Xie X (2019) Neural news recommendation with long-and short-term user representations. In: Proceedings of the 57th annual meeting of the association for computational linguistics. pp. 336–345

Bordes A, Usunier N, Garcia-Duran A, Weston J, Yakhnenko O (2013) Translating embeddings for modeling multi-relational data. Advances in Neural Information Processing Systems 26

Cho K, Van Merriënboer B, Gulcehre C, Bahdanau D, Bougares F, Schwenk H, Bengio Y (2014) Learning phrase representations using rnn encoder-decoder for statistical machine translation. arXiv:1406.1078

Chung J, Gulcehre C, Cho K, Bengio Y (2014) Empirical evaluation of gated recurrent neural networks on sequence modeling. arxiv 2014. arXiv preprint arXiv:1412.3555 1412

Dang CN, Moreno-García MN, Prieta FDL (2021) An approach to integrating sentiment analysis into recommender systems. Sensors 21(16):5666

Ding J, Chen X, Lu P, Yang Z, Li X, Du Y (2023) Dialogueinab: an interaction neural network based on attitudes and behaviors of interlocutors for dialogue emotion recognition. J Supercomput 79(18):20481–20514

Feng C, Khan M, Rahman AU, Ahmad A (2020) News recommendation systems-accomplishments, challenges & future directions. IEEE Access 8:16702–16725

Ge S, Wu C, Wu F, Qi T, Huang Y (2020) Graph enhanced representation learning for news recommendation. Proceedings of the web conference 2020:2863–2869

Ge S, Wu C, Wu F, Qi T, Huang Y (2020) Graph enhanced representation learning for news recommendation. Proceedings of the web conference 2020:2863–2869

Guo H, Tang R, Ye Y, Li Z, He X (2017) Deepfm: a factorization-machine based neural network for ctr prediction. arXiv:1703.04247

Hamborg F, Donnay K, Merlo P et al (2021) Newsmtsc: a dataset for (multi-) target-dependent sentiment classification in political news articles. Association for Computational Linguistics (ACL)

Hu L, Li C, Shi C, Yang C, Shao C (2020) Graph neural news recommendation with long-term and short-term interest modeling. Information Processing & Management 57(2):102142

Hu L, Xu S, Li C, Yang C, Shi C, Duan N, Xie X, Zhou M (2020b) Graph neural news recommendation with unsupervised preference disentanglement. In: Proceedings of the 58th annual meeting of the association for computational linguistics. pp. 4255–4264

Hughes M, Li I, Kotoulas S, Suzumura T (2017) Medical text classification using convolutional neural networks. In: Informatics for health: connected citizen-led wellness and population health. IOS Press, pp. 246–250

Ji Z, Wu M, Yang H, Íñigo JEA (2021) Temporal sensitive heterogeneous graph neural network for news recommendation. Futur Gener Comput Syst 125:324–333

Kipf TN, Welling M (2016) Semi-supervised classification with graph convolutional networks. arXiv:1609.02907

Kozyreva A, Lewandowsky S, Hertwig R (2020) Citizens versus the internet: Confronting digital challenges with cognitive tools. Psychological Science in the Public Interest 21(3):103–156

Leavitt AC (2016) Upvoting the news: breaking news aggregation, crowd collaboration, and algorithm-driven attention on reddit. com. Ph.D. thesis, University of Southern California

LeCun Y, Bengio Y, Hinton G (2015) Deep learning. nature 521(7553):436–444

Li I, Song L, Xu K, Yu D (2022a) Variational graph autoencoding as cheap supervision for amr coreference resolution. In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). pp. 2790–2800

Li, I., Yan, V., Li, T., Qu, R., Radev, D., 2021. Unsupervised cross-domain prerequisite chain learning using variational graph autoencoders. arXiv preprint arXiv:2105.03505

Li J, Zhu J, Bi Q, Cai G, Shang L, Dong Z, Jiang X, Liu Q (2022) Miner: Multi-interest matching network for news recommendation. Findings of the association for computational linguistics: ACL 2022:343–352

Li Y, Tarlow D, Brockschmidt M, Zemel R (2015a) Gated graph sequence neural networks. arXiv:1511.05493

Li Y, Tarlow D, Brockschmidt M, Zemel R (2015b) Gated graph sequence neural networks. arXiv:1511.05493

Loureiro D, Barbieri F, Neves L, Anke LE, Camacho-Collados J (2022) Timelms: Diachronic language models from twitter. arXiv:2202.03829

Mao Z, Li J, Wang H, Zeng X, Wong K-F (2022) Digat: modeling news recommendation with dual-graph interaction. arXiv:2210.05196

Meng X, Huo H, Zhang X, Wang W, Zhu J (2023) A survey of personalized news recommendation. Data Science and Engineering 8(4):396–416

Okura S, Tagami Y, Ono S, Tajima A (2017) Embedding-based news recommendation for millions of users. In: Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining. pp. 1933–1942

Pang B, Lee L et al (2008) Opinion mining and sentiment analysis. Foundations and Trends® in information retrieval 2 (1–2), 1–135

Pennington J, Socher R, Manning CD (2014) Glove: Global vectors for word representation. In: Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP). pp. 1532–1543

Qi T, Wu F, Wu C, Yang P, Yu Y, Xie X, Huang Y (2021) Hierec: Hierarchical user interest modeling for personalized news recommendation. arXiv preprint arXiv:2106.04408

Rendle S (2012) Factorization machines with libfm. ACM Transactions on Intelligent Systems and Technology (TIST) 3(3):1–22

Rieis J, de Souza F, de Melo PV, Prates R, Kwak H, An J (2015) Breaking the news: First impressions matter on online news. Proceedings of the international AAAI conference on web and social media. 9:357–366

Rønningstad E, Velldal E, Øvrelid L (2023) Entity-level sentiment analysis (elsa): An exploratory task survey. arXiv:2304.14241

Sanh V, Debut L, Chaumond J, Wolf T (2019) Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. arXiv preprint arXiv:1910.01108

Shirsat VS, Jagdale RS, Deshmukh S (2017) Document level sentiment analysis from news articles. In: 2017 international conference on computing, Communication, Control and Automation (ICCUBEA). IEEE, pp. 1–4

Tao C, Alatas B (2024a) Real-time emotional topic recommendation in social media news using mdt and hypergraph-based neural networks. IEEE Access

Tao C, Alatas B (2024b) Real-time emotional topic recommendation in social media news using mdt and hypergraph-based neural networks. IEEE Access

Tien JJ, Heng WJ, Fernando ONN (2024) Twittener: Improving news experience with sentiment analysis and trend recommendation. In: International Conference on Human-Computer Interaction. Springer, pp. 417–433

Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser Ł, Polosukhin I (2017) Attention is all you need. Advances in neural information processing systems 30

Veličković P, Cucurull G, Casanova A, Romero A, Lio P, Bengio Y (2017) Graph attention networks. arXiv:1710.10903

Wang H, Wu F, Liu Z, Xie X (2020) Fine-grained interest matching for neural news recommendation. In: Proceedings of the 58th annual meeting of the association for computational linguistics. pp. 836–845

Wang H, Zhang F, Xie X, Guo M (2018) Dkn: Deep knowledge-aware network for news recommendation. In: Proceedings of the 2018 world wide web conference. pp. 1835–1844

Wu C, Wu F, An M, Huang J, Huang Y, Xie X (2019a) Neural news recommendation with attentive multi-view learning. arXiv:1907.05576

Wu C, Wu F, An M, Huang J, Huang Y, Xie X (2019b) Npa: neural news recommendation with personalized attention. In: Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining. pp. 2576–2584

Wu C, Wu F, Ge S, Qi T, Huang Y, Xie X (2019c) Neural news recommendation with multi-head self-attention. In: Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP). pp. 6389–6394

Wu C, Wu F, Huang Y (2021a) Personalized news recommendation: A survey

Wu C, Wu F, Huang Y, Xie X (2021b) User-as-graph: User modeling with heterogeneous graph pooling for news recommendation. In: IJCAI. pp. 1624–1630

Wu C, Wu F, Huang Y, Xie X (2023) Personalized news recommendation: Methods and challenges. ACM Transactions on Information Systems 41(1):1–50

Wu C, Wu F, Qi T, Huang Y (2020) Sentirec: Sentiment diversity-aware neural news recommendation. In: Proceedings of the 1st conference of the Asia-Pacific chapter of the association for computational linguistics and the 10th international joint conference on natural language processing. pp. 44–53

Wu F, Qiao Y, Chen J-H, Wu C, Qi T, Lian J, Liu D, Xie X, Gao J, Wu W et al (2020b) Mind: A large-scale dataset for news recommendation. In: Proceedings of the 58th annual meeting of the association for computational linguistics. pp. 3597–3606

Wu, L., Chen, Y., Shen, K., Guo, X., Gao, H., Li, S., Pei, J., Long, B., et al., 2023b. Graph neural networks for natural language processing: A survey. Foundations and Trends® in Machine Learning 16 (2), 119–328

Xiao T, Han D, He J, Li K-C, de Mello RF (2021) Multi-keyword ranked search based on mapping set matching in cloud ciphertext storage system. Connect Sci 33(1):95–112

Yang B, Liu D, Suzumura T, Dong R, Li I (2023) Going beyond local: Global graph-enhanced personalized news recommendations. In: Proceedings of the 17th ACM conference on recommender systems. pp. 24–34

Yang Z, Wang W, Qi T, Zhang P, Zhang T, Zhang R, Liu J, Huang Y (2024) Glocim: Global-view long chain interest modeling for news recommendation. arXiv:2408.00859

Zhang L, Xu L, Zhang W (2017) Social media as amplification station: Factors that influence the speed of online public response to health emergencies. Asian J Commun 27(3):322–338

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